Machine Learning Model for Predicting Road Network Disruptions During Heavy Precipitation: A Case Study on the Kerala Floods 2018

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Abstract

In the contemporary era, the frequency and severity of global flooding events have alarmingly intensified. A predominant reason behind this escalating trend is human intervention in natural landscapes, resulting in significant modifications to indigenous drainage systems. This, combined with the intensifying ramifications of global climate change, has exacerbated the frequency and intensity of these catastrophic events. One of the most affected facets of urban infrastructure due to these flooding events are the road networks. These networks serve as critical lifelines for transportation and are paramount for swift emergency responses.

The primary focus of this study is to emphasize the pressing need for road networks that are not only robust but also resilient. It is imperative that these networks are designed and maintained with a vision that anticipates potential flood threats, ensuring that they can swiftly adapt to and recover from any disruptions caused by these inundations. To address this challenge, this paper presents a machine learning-driven model that predicts the likelihood of road flooding during episodes of heavy rainfall.

While traditional hydrological models have been cornerstones in flood prediction and analysis, they sometimes grapple with accurately depicting the complex interplay of factors leading to floods. This limitation underscores the potential of machine learning models, which are inherently adept at handling nonlinearities and modelling intricate relationships within vast datasets. Tapping into the power of these advanced algorithms, this research meticulously analyses historical flood data to train and refine the model. The objective is to enhance the model's precision, thereby facilitating accurate flood predictions. This approach is geared towards minimizing the adverse impacts of floods on human life and infrastructural assets.

Acknowledgements

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Declaration

No portion of the work referred to in the dissertation has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Intellectual Property Statement

1. Introduction

1.1 Introduction to the Global Flooding Crisis

Increasing flooding events around the globe have become a growing concern in recent years. Human activities and their impact on natural and hydrological processes are recognized as significant contributors to this trend (Ceola et al., 2014). Urbanization and changes in land use also play a role in the increasing flooding events (Rentschler et al., 2023). The expansion of cities and the conversion of natural landscapes into impervious surfaces, such as concrete and asphalt, reduce the ability of the land to absorb rainfall, leading to increased surface runoff and a higher likelihood of flooding (Tellman et al., 2021; Andreadis et al., 2022). The encroachment of human settlements into flood-prone areas further exacerbates the risk, as more people and infrastructure are exposed to flood hazards (Baldassarre et al., 2010). Climate change is also playing a role, with rising global temperatures leading to more intense rainfall events and an increased likelihood of flooding (Frederikse et al., 2022). Rising global temperatures also increase water vapour in the atmosphere, leading to more intense rainfall and an elevated risk of flooding (Trenberth, 2011).

Furthermore, changes in hydrological systems, such as alterations in river channels and the construction of dams and levees, can impact the natural flow of water and increase the risk of flooding (Lendering et al., 2015). Poor land management practices, such as deforestation and improper soil conservation, can also contribute to increased runoff and soil erosion, further exacerbating the risk of flooding (Madsen et al., 2014).

The impacts of increasing flooding events are significant and wide-ranging. They include damage to infrastructure, loss of lives, displacement of populations, and economic losses (Kundzewicz et al., 2013).

1.2 The Intersection of Floods and Urban Infrastructure

Effects of floods on urban infrastructure can be significant and wide-ranging. Floods can cause damage to buildings, roads, bridges, and other critical infrastructure, leading to disruptions in communication, and essential services (Huong & Pathirana, 2013). The force of floodwaters can erode and weaken structures, compromising their stability and integrity (Hettiarachchi et al., 2018). Floods can also damage electrical systems, water supply networks, and sewage systems, further exacerbating the impact on urban infrastructure (Song et al., 2022).

Furthermore, floods can disrupt the operation of critical infrastructure systems, such as power plants, hospitals, and transportation networks. Power outages can occur due to flooding of electrical substations and damage to power distribution infrastructure (Ramos & Besharat, 2021). Hospitals and healthcare facilities may be forced to evacuate or operate at reduced capacity, impacting the provision of essential medical services (Guan et al., 2021).

Flooding events have significant impacts on roads and transportation networks. During flooding, roads can become submerged or washed away, rendering them impassable and disrupting transportation systems (Pant et al., 2017). The damage to roads can range from surface erosion and potholes to complete structural failure, depending on the intensity and duration of the flood event. Floodwaters can also undermine the stability of road foundations, leading to sinkholes and further compromising the integrity of the transportation network (Pant et al., 2017). Flooded roads and railways can disrupt transportation systems, hindering emergency response efforts and impeding the movement of people and goods (Nasiri et al., 2016).

1.3 Towards Resilient Infrastructure

Hence, the resilience of road networks to flooding is crucial for ensuring the functionality and continuity of transportation systems in urban areas. Resilience refers to the ability of a system to anticipate, absorb, adapt, recover, and learn from disturbances or shocks (Kong et al. 2019). In the context of road networks, resilience entails the capacity to withstand and recover from flood events, minimizing disruptions to transportation services. Resilient road networks are designed and managed to mitigate the impacts of flooding. This includes implementing measures such as proper drainage systems, elevated road sections, and flood-resistant materials in construction (Li et al., 2019). Additionally, advanced modelling and forecasting techniques can help predict flood impacts on road networks, enabling proactive planning and response strategies (Wang et al., 2019) which can help the road networks to adapt to events of shock.

1.4 Leveraging Modern Predictive Techniques

Traditional hydrological models have several limitations in capturing the complex dynamics of flood processes. One limitation is their simplified representation of the hydrological system, which often assumes uniform and homogeneous conditions across the entire watershed (Troy et.al, 2015). This oversimplification may lead to inaccurate predictions, especially in heterogeneous landscapes with varying soil types, land cover, and topography (Chen et al., 2021).

Another limitation is the reliance on historical data and stationary assumptions. Traditional models assume that the statistical properties of hydrological variables, such as rainfall and streamflow, remain constant over time (Read & Vogel, 2015). However, climate change and land use changes can introduce nonstationarity, causing shifts in the frequency and intensity of extreme events (Read & Vogel, 2015). Traditional models may fail to capture these changes, leading to biased predictions and inadequate flood risk assessments.

To overcome these limitations, there is a growing interest in using machine learning (ML) techniques to model flood processes. ML models have been increasingly used in predicting floods due to their advantages in accuracy, efficiency, and convenience (Lu et al., 2022). These models have shown promise in modelling nonlinear relationships and improving forecasting results (Yang & Chang, 2020; Mosavi et al., 2018). Models such as artificial neural networks (ANNs) and support vector machines, have been applied to runoff forecasting and flood prediction (Lu et al., 2022; Mosavi et al., 2018). By analysing historical data and identifying patterns, machine learning algorithms can provide timely forecasts of rainfall and flood hazard variables, aiding in flood preparedness and risk reduction (Chen, 2021).

This study therefore aims at creating a machine learning model that can predict disruptions in road networks during heavy precipitation.

1.5 This Study's Focal Point - The 2018 Kerala Floods

The urgency and significance of this research become even more palpable when considering real-world scenarios, such as the 2018 floods in Kerala, India. With extremely high rainfall of 2346.6 mm from June to August, the rains resulted in severe flooding in 13 districts of Kerala

(Babu & Leno, 2020). The heavy impact of the floods on every sector, including animal husbandry and agriculture, highlights the widespread devastation caused by the event (V.L., 2021; Veerakumaran & Santhi, 2019). The severity of the 2018 floods in Kerala has resulted in extensive damage to roads, making them impassable and affecting the overall connectivity of the region (Mishra et al., 2018; Senan et al., 2022).

1.6 Objective and Significance of the Research

Given these challenges, this project aims to create a model capable of predicting road disruptions during intense precipitation events. The project plans to achieve this through the following objectives:

1. To assess the vulnerability of road networks to flooding, focusing on factors such as elevation, slope, permeability and distance to drainage.

2. To fine-tune a machine learning-based algorithm that can accurately forecast the likelihood of road flooding during heavy precipitation events.

3. To validate the algorithm using historical flood data, particularly from the 2018 floods in Kerala, ensuring its robustness and reliability in real-world scenarios.

Accurate predictions empower authorities and communities to plan and deploy resources efficiently. Also, as someone who has experienced the floods in this study personally, it is not just about maintaining transportation—it is about fortifying the resilience of an entire community against the destructive power of floods.

2 Literature Review

2.1 Importance of Flood Prediction

The importance of flood prediction cannot be overstated, as it plays a crucial role in mitigating the adverse impacts of floods and improving disaster management strategies. Floods are among the most destructive natural disasters, causing extensive damage and loss of life (Mosavi et al., 2018). The costs associated with flood damage and recovery can be substantial, with potential annual losses of up to 9.3% of global gross domestic product (Hinkel et al., 2014). Therefore, accurate flood prediction models are crucial for risk reduction, policy development, and minimizing the loss of human life and property damage associated with floods (Mosavi et al., 2018). Flood prediction also plays a vital role in urban planning and infrastructure development. By accurately predicting flood events, city planners can identify areas at high risk of flooding and implement appropriate measures to mitigate the impact. This may include constructing flood barriers, improving drainage systems, or implementing land-use regulations to prevent construction in flood-prone areas (Danso et al., 2020). Accurate flood prediction can also inform the design and construction of critical infrastructure, such as bridges and dams, ensuring they are built to withstand potential flood events (Jafarzadegan et al., 2023). Furthermore, flood prediction models can help in the management of water resources. Water resource managers can make informed decisions about water allocation and reservoir operations by accurately predicting flood events. This can help prevent overflows and ensure the efficient use of water resources during flood events (Zhang et al., 2018).

2.2 Diverse Approaches to Flood Prediction

Several different flood prediction models are available that utilize various techniques and approaches. These models aim to accurately predict the occurrence and severity of floods to support decision-making and mitigate the impacts of flooding. Some of the notable flood prediction models include:

1. Machine Learning Models: Machine learning techniques, such as artificial neural networks (ANN), support vector regression (SVR), and random forest, have been widely used in flood prediction (Mosavi et al., 2018). These models utilize historical flood data and other relevant variables to train the model and make predictions. They have shown promising results in accuracy and performance (Esfandiari et al., 2020).

2. Hydrological Models: Hydrological models, such as the Hydrologic Engineering Center's Hydrologic Modelling System (HEC-HMS), use mathematical equations to simulate the hydrological processes that lead to flooding (Banihabib et al., 2020). These models predict flood events using rainfall, soil moisture, and river flow. They are commonly used for riverine flood prediction.

3. Statistical Models: Statistical models, such as the index flood method and quantile regression, estimate flood quantiles and predict flood events (Ishak et al., 2011). These models analyse historical flood data and identify patterns and relationships between flood variables. They are often used when at-site recorded flood data are limited.

4. Data-Driven Models: Data-driven models, such as Gaussian processes and deep learning models, utilize large datasets to make predictions (Brockkötter et al., 2021). These models can capture complex relationships and patterns in the data and provide accurate flood predictions (Qian, 2019). They have shown promising results in real-time flood forecasting and early warning systems (Wu et al., 2020).

5. Ensemble Models: Ensemble models combine multiple individual models to improve the accuracy and reliability of flood predictions (Choubin et al., 2019). These models take advantage of the strengths of different models and reduce the uncertainties associated with individual models (Berkhahn et al., 2019). Ensemble models can include a combination of machine learning, hydrological, and statistical models.

6. Remote Sensing Models: Remote sensing data, such as satellite imagery and radar data, can be used in flood prediction models (Schumann et al., 2009). These models utilize the information from remote sensing technologies to estimate flood extent and monitor flood dynamics (Mah et al., 2012). Remote sensing models can provide valuable input data for other flood prediction models.

7. Physics-Based Models: Physics-based models, such as the Saint Venant equations, simulate the physical processes of water flow and flooding (Elhanafy & Copeland, 2007). These models consider the fundamental principles of fluid dynamics and hydraulic behaviour to predict flood events (Lhomme et al., 2008). Physics-based models are often used for detailed flood simulations and floodplain inundation mapping.

2.3 Machine Learning in Flood Prediction

In the realm of flood prediction, various machine learning models have gained prominence. The Support Vector Machine (SVM) stands out for its versatility in handling both classification and regression, finding application in flood susceptibility mapping and forecasting (Li et al.,2019; Liu et al., 2021; Tehrany et al., 2015; Tehrany et al., 2014; Ha et al., 2021). The Random Forest (RF), amalgamates numerous decision trees, enhancing the accuracy of flood predictions (Khalaf et al., 2020; Sampurno et al., 2022; Granata et al., 2016; Li et al., 2016). Artificial Neural Networks (ANN), drawing inspiration from human neural structures, are recognized for their prowess in deciphering intricate nonlinear relationships (Atashi et al., 2022; Tayfur et al., 2018; Hu et al., 2021). Decision trees, with their decision-making tree-like frameworks, have been instrumental in flood forecasting (Choubin et al., 2019; Antonetti et al., 2019). Ensemble models, which merge algorithms like SVM, RF, and ANN, prioritize prediction precision (Jia et al., 2022; Rahman et al., 2021). The nonlinear regression approach, Multivariate Adaptive Regression Splines (MARS), is lauded for addressing complex relationships in flood dynamics (Costache & Bui, 2020). Deep learning models, encompassing Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), have been impactful given their capacity to process voluminous data and identify spatial-temporal flood patterns (Chen et al., 2023; Kondo et al., 2023). Lastly, hybrid models, which merge varied algorithms or couple machine learning with other models like hydrological ones, are tailored to harness the merits of diverse methodologies, aiming for enhanced flood prediction accuracy (Liu et al., 2021; Pham et al., 2017; Xie & Shen, 2021).

2.4 Predictive Features used in ML based Flood Models

In flood prediction model research, a plethora of features have been utilized to optimize their predictive capacities. Meteorological data, encompassing aspects like rainfall patterns and humidity, provides a foundational understanding of precipitation dynamics leading to floods (Ward et al., 2014; Chang et al., 2020). Using high-resolution precipitation data is crucial in flood risk analysis. Precipitation plays a significant role in flooding, and accurate information about its spatial and temporal distribution is essential for understanding flood patterns and assessing flood risk (Shiu et al., 2012).

Complementing this, hydrological variables, such as river flow rates and discharge measurements, elucidate the riverine responses to precipitation events (Tamiru & Wagari, 2021; Tabarestani & Afzalimehr, 2021). The terrain's influence on flood dynamics is assessed through topographic data, highlighting elevation and slope gradient factors that direct water flow (Li et al., 2021; Wang et al., 2021). By analysing the slope of roads derived from Digital Elevation Models (DEM), we can identify areas prone to water accumulation and potential drainage issues during flood events (Ferencevic & Ashmore, 2011). The elevation information extracted from DEMs allows for the identification of low-lying areas and areas at risk of inundation during floods (Hashemi-Beni et al., 2018). This information is essential for understanding the spatial extent and severity of flood events and for developing effective flood risk management strategies (Hashemi-Beni et al., 2018).

Land use and land cover data, which details the spatial arrangement of urban areas, forests, and other terrains, further influences flood genesis by affecting infiltration and runoff processes (Li et al., 2021). Soil properties, including moisture content and infiltration capacity, contribute to runoff generation dynamics (Wang et al., 2021; Tabarestani & Afzalimehr, 2021).

One study conducted in Shanghai, China, evaluated the impact and risk of pluvial flash floods on the intra-urban road network (Yin et al., 2016). The results showed that roads closer to drainage networks experienced a greater impact from flooding. This finding is consistent with studies conducted in other cities, such as Portland and Boston, where urban transportation networks also demonstrated a higher vulnerability to flooding (Yin et al., 2016).

A study focused on the variation of surface runoff and soil moisture content in the subgrade of permeable pavement structures (Hou et al., 2020) found that permeable roads can greatly alleviate the pressure of urban drainage and reduce the risk of storms and floods.

Advancements in technology, notably remote sensing, offer insights on vegetation and surface water distribution, enhancing the spatial comprehension of flood variables (Munawar et al., 2022). Historical flood data, encompassing prior flood locations and characteristics, serves as an essential reference to discern patterns and relationships (Mistry & Parekh, 2022; Tabbussum & Dar, 2021). Lastly, other contingent factors such as anthropogenic influences and dam operations might be factored into models depending on the specific research context (Ward et al., 2014; Tabbussum & Dar, 2021).

2.5 Data Imbalance and its Rectification

The reality of most natural disaster datasets is their imbalance. The dominance of "non-event" data (in this case, non-flooded roads) can skew model training, leading to biased predictions (Rahman & Davis, 2013). This is because the models are more likely to be biased towards the majority class, as they have more examples to learn from (Chawla et al., 2002). As a result, the minority class may be overlooked or misclassified, leading to poor performance in real-world applications (ŞİMŞEK & DAŞ, 2022).

To address the challenges posed by class imbalance, researchers have proposed various techniques. One approach is to use sampling methods, such as oversampling the minority class or undersampling the majority class, to balance the class distribution (Chawla et al., 2002). Oversampling techniques, such as Synthetic Minority Over-sampling Technique (SMOTE), generate synthetic examples of the minority class to increase its representation in the dataset (Chawla et al., 2002). Undersampling techniques, on the other hand, reduce the number of examples from the majority class to match the number of examples in the minority class (Bach et al., 2017). These techniques aim to create a more balanced training set, allowing the model to learn from both classes more effectively.

2.6 Advantages of ML Based Models

Machine learning models present a range of superiorities in flood prediction compared to conventional techniques. Firstly, they excel in discerning complex relationships and pinpointing intricate patterns and nonlinear interactions, which standard models might often miss (Atashi et al., 2022). Anchored in a data-driven paradigm, they harness extensive historical flood records, enabling them to refine and enhance predictions iteratively. This adaptability extends to diverse flood types and geographic areas, and by tailoring them to specific datasets, they can be aptly aligned with a region's unique attributes (Kohansarbaz et al., 2022). An integrative capability allows these models to amalgamate data from diverse avenues like remote sensing, meteorological records, and past flood events (Sankaranarayanan et al., 2019). Addressing one of the recurrent challenges, these models capably manage missing or fragmentary data, ensuring consistent forecasting despite data gaps (Tayfur et al., 2018). Emphasizing their adaptability, they are scalable, proficiently handling extensive tasks, a feature pivotal for prominent urban localities or expansive river catchments (Kondo et al., 2023).

2.7 Kerala Floods 2018: A Case Study

Kerala, located in India, has consistently faced floods over the years. A particularly severe flood event hit the region in August 2018, affecting around 23 million individuals. This disaster was regarded as Kerala's most detrimental since the Great flood of 1924 (Divakaran et al., 2019). The flood's intensity resulted from a combination of substantial rainfall and the release of water from several dams, leading to vast destruction across infrastructure, agriculture, and many people's livelihoods (Kumar et al., 2020). Notably, the socio-economic well-being of dairy farmers was significantly disrupted by this flood event (V.L., 2021).

Research suggests that the extreme precipitation and flooding in Kerala in 2018 were influenced by climate change (Dhasmana et al., 2023). Climate models indicate a significant increase in rainfall and moisture flux over the peninsula's southern part, contributing to the flooding (Hunt & Menon, 2020). Another factor contributing to the floods was changing land use and land cover (LULC) (Dixit et al., 2022). Improper dam operations also affected the flooding (Dixit et al., 2022). The opening of dams to release excess water exacerbated the flooding in the region (Neeraj et al., 2020).

The effects of the flood on the road infrastructure were severe. The floodwaters washed away roads and caused landslides, making many roads impassable (Thakur et al., 2020). This significantly impacted the transportation network in the affected areas, hindering the movement of people and goods. The flood also had long-term effects on the road infrastructure. The damage caused by the flood required extensive repairs and reconstruction of roads and bridges. The cost of repairing the road infrastructure in Kerala after the flood was estimated to be substantial (Mishra et al., 2018).

In the subsequent years, Kerala confronted back-to-back flood challenges, especially in 2018 and 2019 (Joseph, 2022). Such events accentuated the requirement for strengthened disaster management strategies within the state. There have been discrepancies between predicted flood-vulnerable zones and the actual regions that got affected, suggesting a pressing need to integrate effective disaster mitigation plans in Kerala's Municipal Corporations' planning documentation (Shankar & Bindu, 2021).

Multiple research initiatives have been undertaken to gain insights into the root causes of Kerala's floods and to propose potential countermeasures. Kumar et al. (2020) dissected the role of meteorological elements, such as low-pressure systems and dry air intrusions, in exacerbating the August 2018 flood. Their findings suggested that both heavy rainfall and dam water releases intensified the flooding. Furthermore, a different study concentrated on the aftermath of climate change on Kerala's public health infrastructure during these floods, underlining Kerala's susceptibility to climate-affected calamities (Varughese & Purushothaman, 2021). Asim et al. (2019) shed light on the mental health of the affected, noting a significant prevalence of post-traumatic stress disorder (PTSD) among the flood victims. This is further echoed by Jose & Fenn (2021) who discussed the profound psychological implications on farmers, underscoring the role of resilience in their recovery journey. Divakaran et al. (2019) ventured into the biological aftermath of the flood, discovering intricate bacterial compositions and antibiotic resistance patterns in the inundated areas. In pursuit of a comprehensive flood hazard analysis, Thakur et al. (2020) synergized remote sensing, GIS, and hydrological models. Dhasmana et al. (2023) examined the flood through the lens of climate change, providing evidence of its influence on the disaster's severity and highlighting the urgency for adaptation strategies. Urban preparedness surfaced as a theme in Shankar & Bindu (2021), emphasizing the imperative to embed disaster mitigation into municipal planning. With public health repercussions at the forefront, Varughese & Purushothaman (2021) spotlighted the vulnerability of Kerala's health systems in the face of such climate-induced adversities. Exploring the economic realm, V.L., (2021) focused on the socio-economic repercussions on dairy farmers, while Greeshma & Greeshma (2023) assessed the economic tremors felt by the plantation sector. George et al. (2022) tapped into the potential of the Hydrologic Engineering Centers Hydrologic Modelling System (HEC-HMS), illustrating its efficacy in flood modelling. Meanwhile, Veerakumaran & Santhi (2019) presented a narrative of the flood's impact on agriculture, and Pradeep & Rajesh (2020) narrated the socio-economic disruptions endured by communities. Post-flood, Babu &

Leno (2020) delved into the environmental facet, assessing the organic carbon status in flood-impacted terrains.

Post the 2018 floods, research on Kerala's rehabilitation process indicated both successes and challenges in implementing the "Build Back Better" (BBB) principle in recovery initiatives (Neeraj et al., 2020). This underscores the significance of adopting an all-encompassing and enduring post-flood recovery framework.

During these floods, social media emerged as a crucial tool in organizing and facilitating disaster relief efforts, aiding in resource allocation, unifying relief activities, and voicing the urgency for rescue missions (Joseph, 2022).

Lastly, to robustly counter flood-related challenges, Kerala needs to embrace a holistic flood mitigation plan comprising well-strategized land utilization, water regulation, and infrastructural advancements. Given that elements like changes in land usage, enhanced impermeable areas, and evolving hydrological scenarios contribute to Kerala's flood vulnerabilities, it's imperative to incorporate them in flood risk evaluation and counter-strategies (Mishra et al., 2018; Sarkar & Mondal, 2019; Dang & Kumar, 2017).

3. Methodology

3.1 Choice of Study Area: Ernakulam and Idukki Districts, Kerala

Kerala, a state located in the southwestern region of India, is characterized by various geographical features. The state is relatively small in size and is situated on the southern peninsular region of India (Ramanarayanan & Rajeev, 2020). It is bordered by the Arabian Sea on the west and is known for its beautiful coastline (Hunt & Menon, 2020). The state is also home to the Western Ghats, a mountain range that runs parallel to the Arabian Sea and is known for its rich biodiversity and scenic landscapes (Hunt & Menon, 2020). The Western Ghats play a crucial role in the monsoon rainfall patterns of Kerala (Simon & Mohankumar, 2004).

Kerala experiences a tropical climate, with high temperatures and humidity throughout the year (Lu et al., 2020). The state receives heavy rainfall during the monsoon season, which lasts from June to September (Ananthakrishnan & Soman, 1988). The monsoon rains are vital for the agriculture and ecosystem of the region. Kerala's rivers, such as the Periyar, Pamba, and Bharathapuzha, are fed by these monsoon rains and play a significant role in the state's irrigation and water supply (Simon & Mohankumar, 2004).

The coastal plains of Kerala are characterized by backwaters, which are a network of interconnected canals, lakes, and lagoons. These backwaters are not only a unique geographical feature but also a major tourist attraction, known for their scenic beauty and houseboat cruising. The state is also known for its sandy beaches, such as Kovalam and Varkala, which attract tourists from around the world (Hunt & Menon, 2020).

In addition to its coastal and mountainous regions, Kerala is also known for its rich biodiversity. The state is home to numerous wildlife sanctuaries and national parks, including Periyar Tiger Reserve, Wayanad Wildlife Sanctuary, and Silent Valley National Park (Hunt & Menon, 2020). These protected areas are crucial for the conservation of various species of flora and fauna. The study area map is shown in Figure 1.

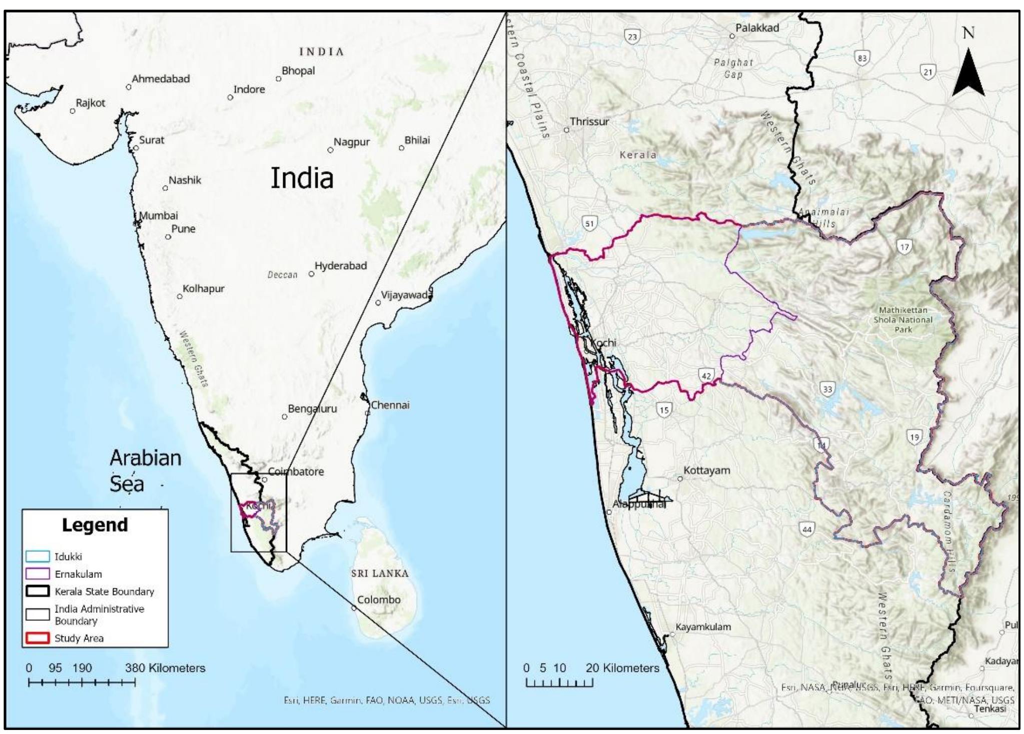

Figure 1: The study area comprising the districts of Ernakulam and Idukki in the state of Kerala, India. The basemap is the courtesy of ESRI 2023.

The state witnessed unprecedented floods in the year 2018 which primarily affected the central parts of the state, mainly the districts of Ernakulam and Idukki, and resulted in the inundation of approximately 1100 km^2 of land (Vishnu et al., 2019). The flood was caused by a combination of extreme rainfall and the release of water from reservoirs in the region (Mishra et al., 2018). In Ernakulam district, which is located on the coastal plains of Kerala, the heavy rainfall and rising water levels resulted in the submergence of low-lying areas, including roads and bridges, leading to disruptions in transportation (Vishnu et al., 2019). Similarly, in Idukki district, which is situated in the mountainous region of the Western Ghats, the floods caused significant damage to the road infrastructure (Vishnu et al., 2019). The heavy rainfall and landslides in the hilly terrain resulted in the blockage and collapse of roads, making them impassable (Vishnu et al., 2019). Intense rainfall events, such as the 177.5 mm of rainfall on August 17, 2018, in Idukki, led to overflowing reservoirs and dams in the highland and midland areas (Vishnu et al., 2019).

The significance of using the 2018 Kerala floods as a study area for creating a model that can predict road flooding during heavy precipitation lies in the severity and complexity of the event. By studying this event, researchers can gain valuable insights into the factors that contribute to the severity of flooding and develop a predictive model to anticipate and mitigate the impacts of future flooding events on road infrastructure. Kerala's vulnerability to extreme weather events, its high rainfall, and its complex topography make it an ideal study area for understanding the dynamics of road flooding and developing effective strategies for disaster management and urban planning (Singh et al., 2018). The insights from studying the 2018 Kerala floods can also be applied to other regions facing similar challenges, where heavy precipitation and inadequate infrastructure can lead to road flooding and associated disruptions (Ginkel et al., 2021). By developing a model that can accurately predict road flooding during heavy precipitation, authorities can improve their preparedness and response strategies, ultimately reducing the loss of life, property, and economic costs associated with road infrastructure damage during floods (Lamb et al., 2019).

3.2 Data Acquisition

Data acquisition for flood risk analysis in developing nations like India poses several challenges. One of the main challenges is the need for more accurate and reliable data (Mansour, 2020). This is crucial for effective flood risk management and the development of appropriate mitigation strategies. The availability of accurate data is essential for identifying flood-prone areas, assessing vulnerability, and understanding the potential impacts of flooding (Mansour, 2020). This was the major challenge faced in the data collection stage of this research, as data about the 2018 floods were scarcely available. Based on literature, it was decided that the training of the disruption likelihood model requires data about the flooded roads and their characteristics like elevation, slope, permeability, precipitation and distance to drainage, to predict the availability of the road for use. Since an exhaustive list of this kind of data was unavailable, this research generated the data from remote sensing products.

Utilizing geospatial data files, we can access intricate details about road networks, which proves instrumental in assessing potential risks and impacts of floods on urban road systems (Yin et al., 2016). Such files, when incorporated into flood modelling, enable a systematic evaluation of the effects of flash flood events on these networks (Yin et al., 2016). Since there was an unavailability of information about the affected road networks in the study area, this research aimed at using shapefiles of the road networks so that they could be further used to extract characteristics of each of the roads in the study area. The road network shapefiles required for this work was obtained from Geofabrik. Geofabrik is a server that provides data extracts from the OpenStreetMap (OSM) project (Haklay & Weber, 2008). OSM is a collaborative mapping project that allows users to contribute and edit street maps (Haklay & Weber, 2008). Geofabrik's server offers free downloads of OSM data extracts, which are regularly updated (Abshirini & Koch, 2016). The data downloaded included shapefiles of places, land use, transport, buildings etc. The road networks shapefile was extracted from this data and was then clipped to the study area extent as shown in Figure 2 below.

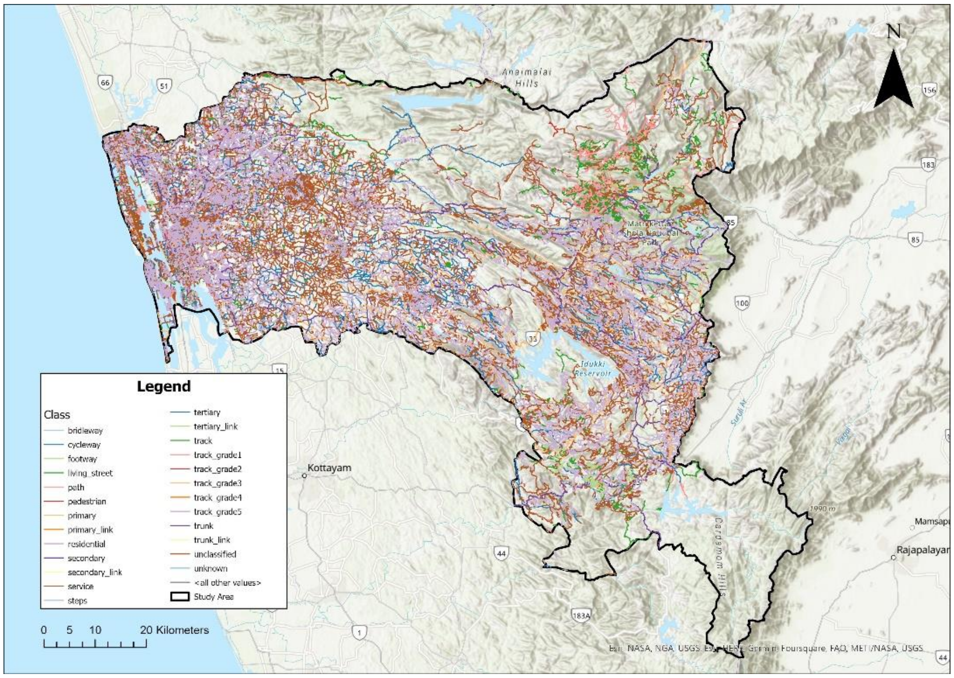

Figure 2: Road networks in the study area divided into different classes. The basemap is the courtesy of ESRI 2023.

To acquire information about the elevation and slope of the road networks, a DEM provide by the Shuttle Radar Topography Mission (SRTM), acquired from the OpenTopography portal was used (Figure 3). The SRTM was a space mission conducted by NASA and the National Imagery and Mapping Agency (NIMA) with the goal of creating a high-resolution DEM of the Earth's surface (Farr et al., 2007). The mission was flown in February 2000 and collected data using a radar instrument onboard the Space Shuttle Endeavour (Zandbergen, 2008). The SRTM mission aimed to provide accurate and consistent elevation data on a global scale, covering approximately 80% of the Earth's surface between 60°N and 56°S (Zandbergen, 2008).

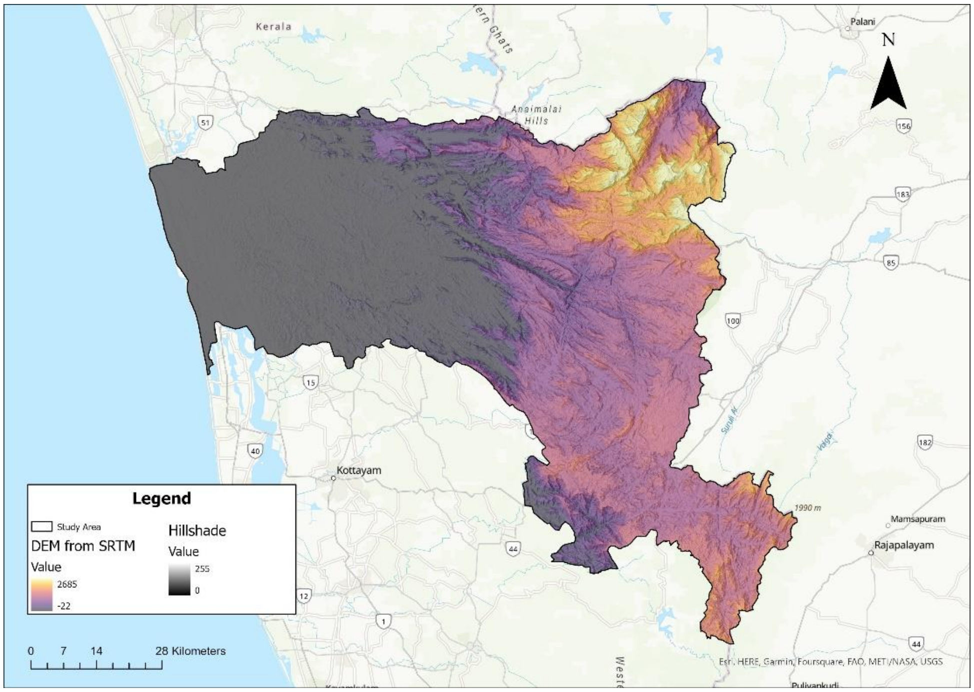

Figure 3: Digital Elevation Model of the study area obtained from SRTM.
The basemap is the courtesy of ESRI 2023.

The highest resolution precipitation data available for the study area was from Climate Hazards InfraRed Precipitation with Station data (CHIRPS). CHIRPS is a satellite-based precipitation dataset that provides high-resolution and long-term estimates of rainfall (Funk et al., 2015). It combines infrared Cold Cloud Duration (CCD) observations with smart interpolation techniques to create a global record of precipitation (Funk et al., 2015). The CHIRPS dataset has relatively high spatial and temporal resolutions, making it suitable for various applications, including drought monitoring, hydrological modelling, and climate change studies (Dinku et al., 2018; Bayissa et al., 2017). The resolution of the data was about 6 km. Even though this is higher than the resolution of the DEM used in this study, this raster was able to give some reliable information about the variability of precipitation in the study area.

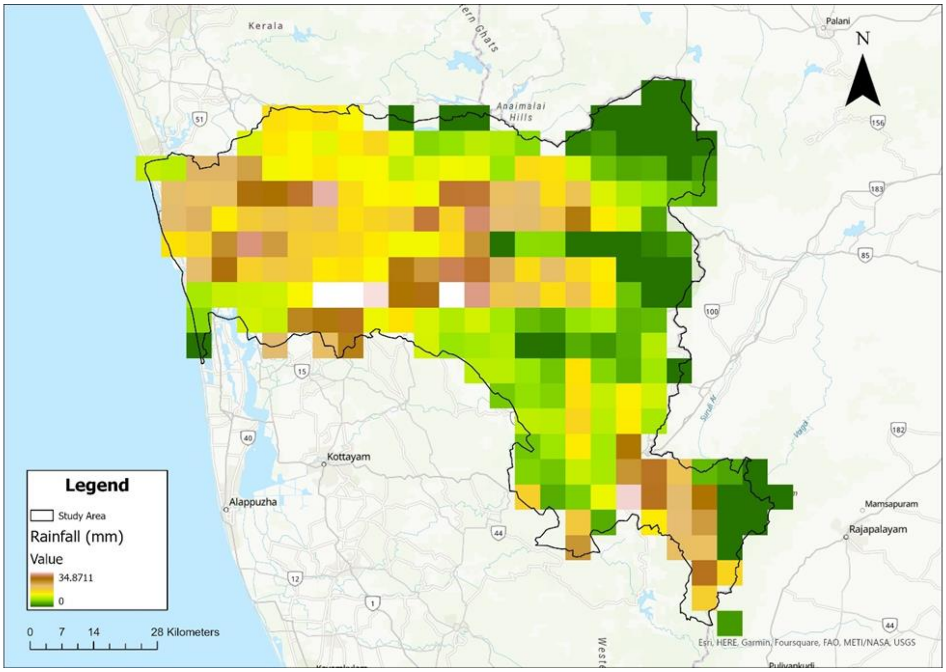

Figure 4: Precipitaion raster of the study area. The basemap is the courtesy of ESRI 2023.

To acquire data about the distance to the closest drainage network for each road, Geofabrik server was used. The shapefiles acquired of the drainage network is shown in Figure 5.

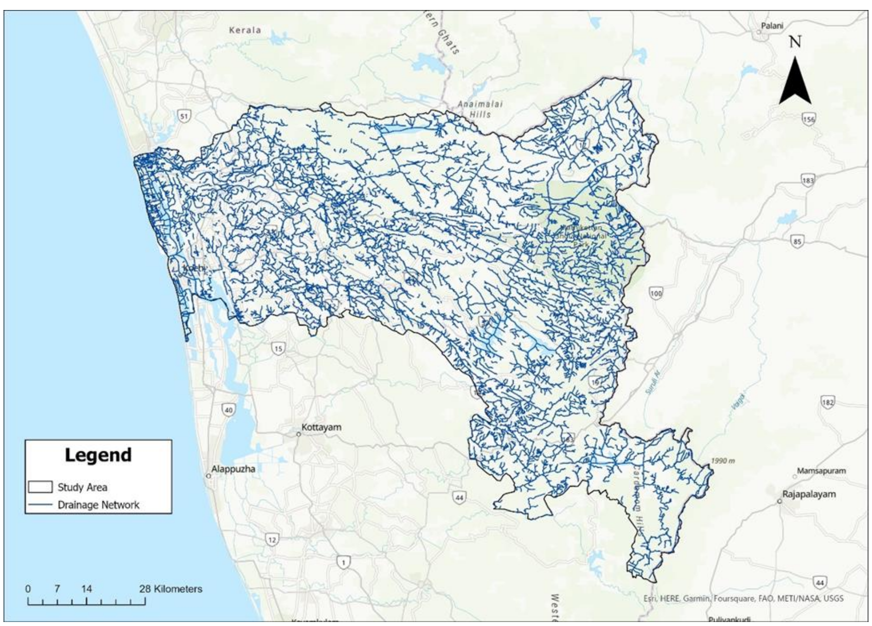

Figure 5: Drainage networks in the study area. The basemap is the courtesy of ESRI 2023.

A raster layer created by Tsinghua University was acquired from GEE which was then used to identify how permeable each road network was. This dataset contains annual change information of global impervious surface area from 1985 to 2018 at a 30m resolution (Gong et al., 2020)

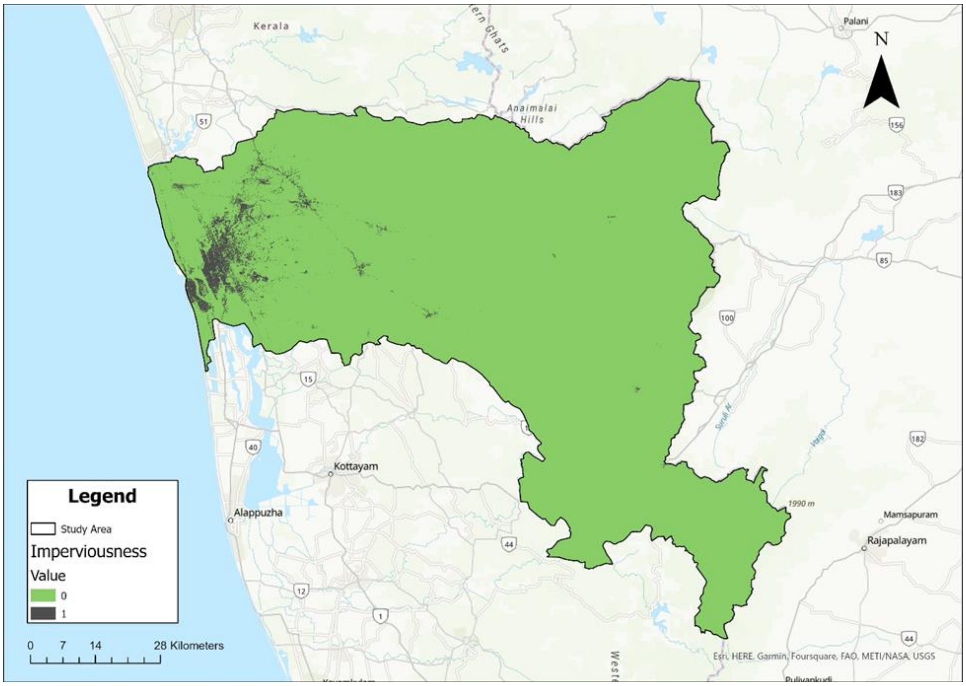

Figure 6: Raster showing impervious pixels in the study area. The basemap is the courtesy of ESRI 2023.

Finally, data about which roads were flooded during the 2018 event was required so that it could be used to train the model. The first attempt made was to acquire Landsat images from Google Earth Engine (GEE) to identify flooded regions and then intersect them with the roads to see which of them were flooded. But the challenge here was that almost all the images from this period had very high cloud cover which made it impossible to identify the flooded regions. This was a limitation of all the images acquired from an optical sensor. This is when an approach was made to use Synthetic Aperture Radar (SAR) images. SAR images have the ability to penetrate clouds and heavy rainfall, making them particularly valuable for flood mapping and evaluation (Huang & Jin, 2020). So, a SAR image was acquired for the date of 21st August 2018 from ESA Copernicus which was then pre-processed in SNAP software. Then a global thresholding approach was used to identify the water pixels (Figure 7) from the image which was then used to identify the flooded roads.

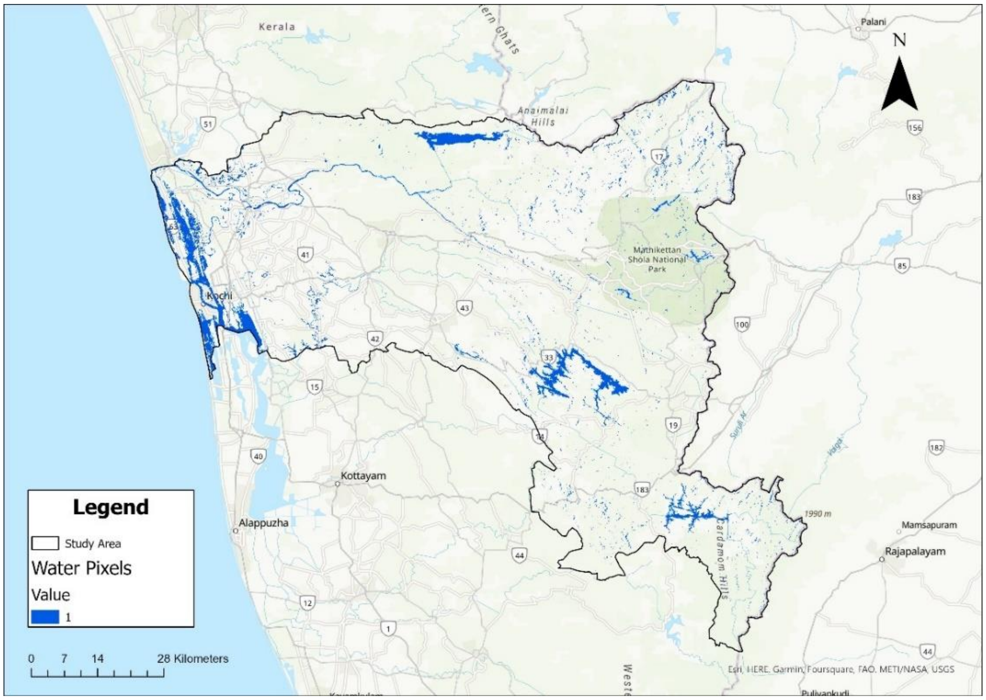

Figure 7: Raster showing water pixels in the study area. The basemap is the courtesy of ESRI 2023.

3.3 Data Integration & Feature Extraction

The zonal statistics as table tool in ArcGIS Pro was used to extract the elevation information of the roads. The input feature class or zones was selected as the road networks shapefile and the raster was selected as the DEM. Each road has a unique osm_id which can be used to identify the road. This was used as the zone field which would be used to calculate the zonal statistics. This tool was used to extract the mean, minimum and maximum elevation of each road network.

The slope tool in ArcGIS Pro was then used to create raster layer from the dem with each pixel showing the slope. The zonal statistics as table tool was then uses to extract the minimum, maximum and mean slope of each of the roads.

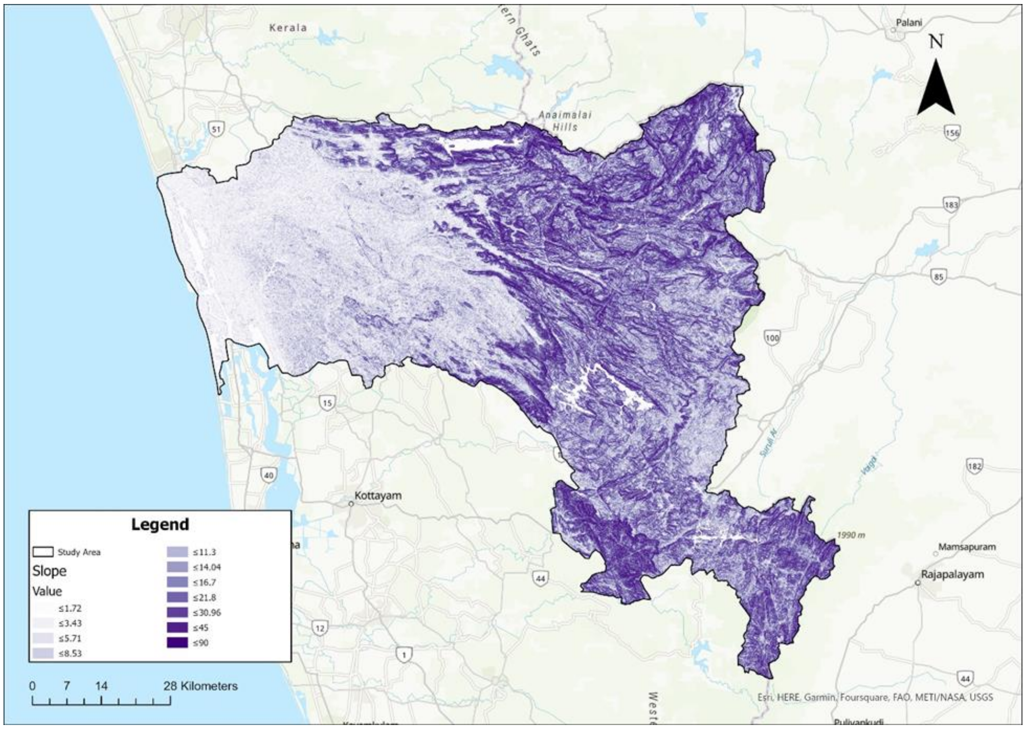

Figure 8: Slope raster obtained from the DEM of the study area. The basemap is the courtesy of ESRI 2023.

The imperviousness raster obtained from GEE contained information on when each pixel in the raster became impervious. This raster was reclassified so that all the impervious pixels had a value of 1 and the rest had value of 0. This was done so that when the zonal statistics as table tool would be used to get the sum of these pixels that come under each road, a higher value would mean that a large region around the road is impervious.

Even though the study area is only two districts from the state of Kerala, there can be variations in the amount of precipitation. To obtain information about the precipitation that each road received, the zonal statistics as table tool was again used with the roads as the zones and the precipitation layer as the input raster.

The drainage network data obtained from OSM was then used with the near tool in ArcGIS Pro to identify the distance from each road to its nearest drainage source.

The zonal statistics as table tool was used on the water pixels layer created in SNAP to identify flooded roads. The tool was used to calculate the sum of pixel values under each road. If the sum came out to be 0, it would mean that there are no water pixels under the road and hence the road was not flooded.

For each road in the network, characteristics like mean elevation, minimum elevation, maximum elevation, mean slope, minimum slope, maximum slope, mean precipitation, maximum precipitation, minimum precipitation, imperviousness and distance to drainage were compiled a csv file. It was found that there was a very big imbalance in the data. To rectify this, the oversampling method of SMOTE was employed. The advantage of using SMOTE is that it does not result in a loss of useful data (Shi et al., 2022). Unlike undersampling methods that discard instances from the majority class, SMOTE creates synthetic examples by interpolating between existing minority class instances (Shi et al., 2022). This ensures that all available information is utilized in the training process.

After balanced data was ensured, to find out the features that were significant in predicting flood, Recursive Feature Extraction (RFE) and statistics from a logistic regression model were used. Ding & Wilkins (2006) highlight that RFE is a well-studied method for reducing the number of attributes used in prediction models or further analysis. By iteratively eliminating features based on their importance, this technique helps to identify the most relevant features and improve the efficiency and interpretability of the models. RFE was implemented through the python script shown in Figure 9.

```
data_final_vars=os_data_X.columns.values.tolist()
y=os_data_y['flooded']
X=[i for i in data_final_vars if i not in y]
from sklearn.feature_selection import RFE
from sklearn.linear_model import LogisticRegression
logreg = LogisticRegression(C= 0.5, solver='saga',)
rfe = RFE(estimator=logreg, n_features_to_select=5)
rfe = rfe.fit(os_data_X, os_data_y.values.ravel())
print(rfe.support_)
print(rfe.ranking_)

selected_features = [X[i] for i, x in enumerate(rfe.support_) if x]
print(selected_features)
```

Figure 9: Snippet of Python script for RFE.

The significance of logistic regression coefficients can be assessed through hypothesis testing. The p-values associated with the coefficients indicate whether the relationship between each predictor and the outcome is statistically significant. A low p-value (typically below a predetermined significance level, e.g., 0.05) suggests that the coefficient is significantly different from zero, indicating a meaningful relationship (Kim et al., 2014). This was implemented in this study using statsmodels in python as shown in Figure 10.

```
import statsmodels.api as sm
logit_model=sm.Logit(os_data_y,os_data_X)
result=logit_model.fit()
print(result.summary2())
```

Figure 10: Snippet of Python script for getting statistics from the Logistic Regression Model.

The features that were not significant were removed from the data frame based on the results from these two implementations and the rest was used to train the model. This data was used in python using statsmodels to perform logistic regression. This package provides detailed statistical summaries, which are often useful for understanding the significance and fit of your model

3.4 Model Selection and Rationale

This study has used a Logistic Regression Model. Logistic regression is a widely used statistical modelling technique that is suitable for binary classification problems, such as predicting flood occurrence or non-occurrence. It provides interpretable results by estimating the probability of an event based on input variables Wei et al. (2009). Logistic regression has been applied in flood susceptibility mapping, flood stage forecasting, and flood risk assessment (Tehrany et al., 2014; Giovannettone et al., 2018). It is computationally efficient and can handle large datasets, making it a practical choice for flood prediction models (Rahmati et al., 2020). Logistic regression can also provide insights into the relative importance of input variables in influencing flood occurrence (Giovannettone et al., 2018).

Logistic regression provides interpretable results and is computationally efficient, but it assumes a linear relationship between input variables and flood occurrence (Yaseen et al., 2022).

3.5 Model Validation and Testing

Once models are trained, it's imperative to validate their predictions on unseen data. This study has used a k-fold cross validation technique to assess the performance and generalisation of the model. It involves dividing the available data into k subsets or folds, using k-1 folds for training the model and the remaining fold for testing or validation (Li, 2023). The process is repeated k times, with each fold serving as the validation set once. One of the key advantages of k-fold cross-validation is that it provides a more robust estimate of model performance compared to a single train-test split (Fushiki, 2009). By repeating the process k times and averaging the results, it reduces the impact of the specific data partitioning on the evaluation metrics. This helps to mitigate the risk of overfitting or underfitting that can occur with a single train-test split.

The whole methodology summarised into a flow chart is shown in Figure 11.

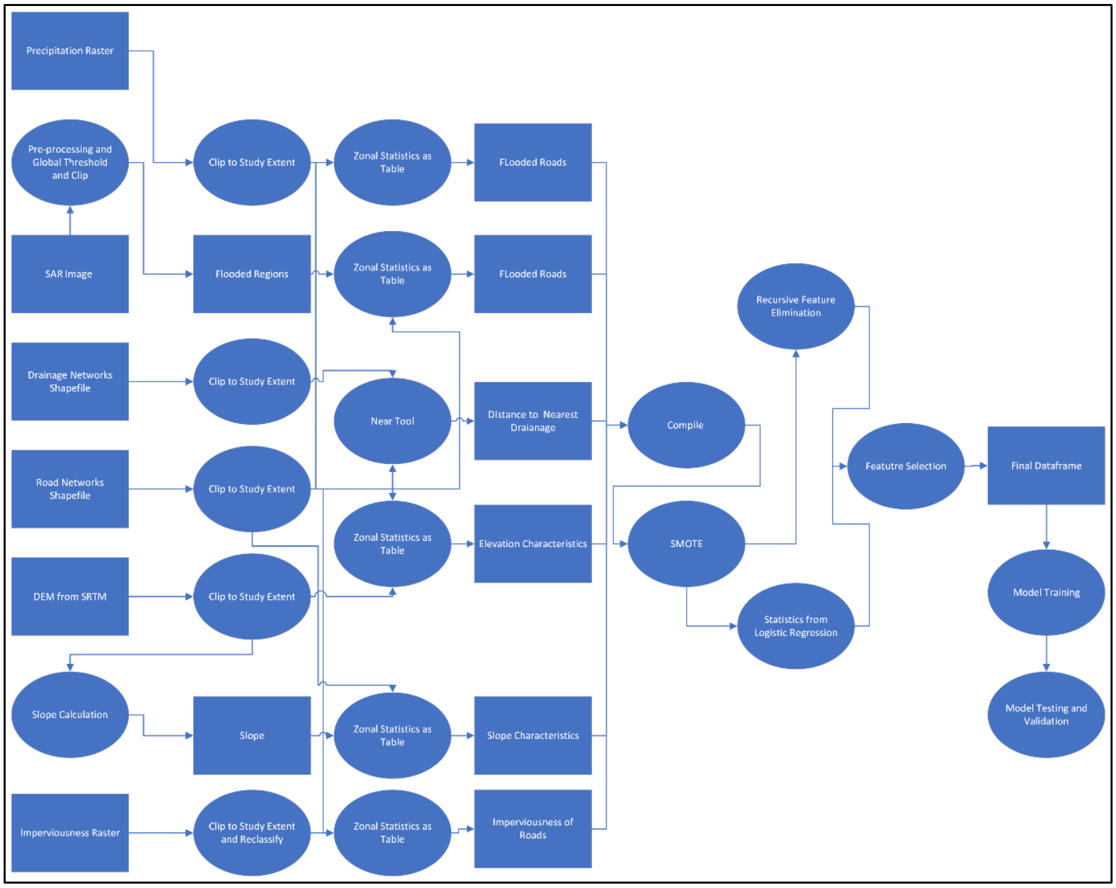

Figure 11: Flow chart showing the methodology.

4. Results

4.1 Feature Extraction

There are a total of 68,404 roads in the study area which are classified into primary, secondary, tertiary, etc. as given in Table 1.

Table 1: The different classes of roads in the dataset.

Class	Count	Class	Count
bridleway	8	tertiary	2926
cycleway	29	tertiary_link	56
footway	812	track	2703
living_street	400	track_grade1	18
path	2847	track_grade2	20

pedestrian	92	track_grade3	17
primary	1384	track_grade4	15
primary_link	98	track_grade5	10
residential	33194	trunk	615
secondary	1092	trunk_link	154
secondary_link	55	unclassified	18342
service	3390	unknown	52
steps	75	**Total**	68404

Out of the total number of roads, only 765 were flooded and the rest 67639 were not flooded. The statistics of the different characteristics of each of the road is shown in Tables 2 and 3.

Table 2: Summary statistics of min_elevation, max_elevation, min_slope, min_precipitation and max_precipitation.

	mean_elevation	mean_slope	mean_precipitation	imperviousness	flooded	distance_to_drainage
count	68404.00	68404.00	68404.00	68404.00	68404.00	68404.00
mean	309.40	10.65	18.43	4.92	0.01	258.10
std	520.33	9.90	7.42	15.30	0.11	281.72
min	-3.00	0.00	0.00	0.00	0.00	0.00
25%	10.50	4.12	13.25	0.00	0.00	31.80
50%	24.83	6.93	18.18	0.00	0.00	175.86
75%	557.00	13.37	23.24	4.00	0.00	385.37
max	2517.41	218.46	34.87	527.00	1.00	2388.40

Table 3: Summary statistics of mean_elevation, mean_slope, mean_precipitation, imperviousness, and distance_to_drainage.

min_elevation	max_elevation	min_slope	max_slope	min_precipitation	max_precipitation
68404.00	68404.00	68404.00	68404.00	68404.00	68404.00
301.83	316.75	5.42	16.63	18.27	18.60
514.97	525.99	7.20	15.22	7.47	7.48
-8.00	-3.00	0.00	0.00	0.00	0.00
9.00	12.00	1.28	6.41	13.21	13.30
21.00	29.00	2.87	11.07	18.18	18.18
546.00	566.00	6.31	21.22	23.24	24.31
2472.00	2621.00	163.40	443.01	34.87	34.87

4.2 Logistic Regression Statistics for Feature Selection

The summary statistics from the Logistic Regression model is shown in Figure 12.

```
Optimization terminated successfully.
         Current function value: 0.066212
         Iterations 10
                          Results: Logit
======================================================================
Model:                Logit              Method:              MLE
Dependent Variable:   flooded            Pseudo R-squared:    -0.079
Date:                 2023-10-10 12:58   AIC:                 9080.3377
No. Observations:     68404              BIC:                 9180.8027
Df Model:             10                 Log-Likelihood:      -4529.2
Df Residuals:         68393              LL-Null:             -4198.1
Converged:            1.0000             LLR p-value:         1.0000
No. Iterations:       10.0000            Scale:               1.0000
----------------------------------------------------------------------
                       Coef.  Std.Err.    z     P>|z|   [0.025  0.975]
----------------------------------------------------------------------
mean_elevation         0.0021   0.0039   0.5263 0.5986 -0.0056  0.0098
mean_slope            -0.0298   0.0127  -2.3460 0.0190 -0.0548 -0.0049
mean_precipitation     0.1261   0.2322   0.5433 0.5869 -0.3289  0.5811
imperviousness        -0.0347   0.0062  -5.6402 0.0000 -0.0468 -0.0227
distance_to_drainage  -0.0034   0.0002 -14.7751 0.0000 -0.0038 -0.0029
min_elevation         -0.0068   0.0019  -3.6613 0.0003 -0.0105 -0.0032
max_elevation          0.0026   0.0023   1.1257 0.2603 -0.0019  0.0070
min_slope             -0.0506   0.0111  -4.5526 0.0000 -0.0724 -0.0288
max_slope              0.0345   0.0057   6.0305 0.0000  0.0233  0.0458
min_precipitation     -0.1790   0.1133  -1.5802 0.1141 -0.4011  0.0430
max_precipitation     -0.1542   0.1204  -1.2800 0.2006 -0.3902  0.0819
======================================================================
```

Figure 12: Summary statistics from the Logistic Regression model used for feature selection.

Upon evaluating the logistic regression model for the 'flooded' outcome, it was discerned that the model optimally converged after 10 iterations. In terms of model fit, the pseudo R-squared value stood at -0.079, not directly indicative of variance explained like in linear regression, and possibly suggesting a fit no better than a model with just a constant. This is further accentuated by its log-likelihood value of -4529.2, and an LLR p-value of 1.0000, hinting that the model's statistical superiority over a rudimentary, intercept-only model might be questionable.

Interpreting Significant Coefficients:

Mean slope: For every unit increment in mean_slope, the log-odds of 'flooded' decrease by 0.0298. This observation is statistically significant ($p < 0.05$).

Imperviousness: A unit rise in imperviousness causes the log-odds of 'flooded' to decrease by 0.0347.

Distance to drainage: A single unit increase corresponds to a reduction in the log-odds of 'flooded' by 0.0034, a statistically crucial correlation.

Minimum elevation: Each unit enhancement in min_elevation leads to a decline in the log-odds of 'flooded' by 0.0068.

Minimum slope: An increment in min_slope reduces the log-odds of 'flooded' by 0.0506.

Maximum slope: An increase in max_slope augments the log-odds by 0.0345, showing its significance in the model.

Potentially Insignificant Predictors:

Certain variables like mean_elevation, mean_precipitation, max_elevation, min_precipitation, and max_precipitation present p-values exceeding the 0.05 threshold, hinting that their significance in predicting the 'flooded' outcome might be questionable in this model's context.

4.3 Recursive Feature Elimination Results

Recursive feature elimination was also used on the same database to identify the 5 most significant features. The results after running the script is shown in Figure 13. These results can be interpreted as shown in Table 4.

```
[False  True  True  True False False False  True  True False False]
[5 1 1 1 7 6 4 1 1 2 3]
['mean_slope', 'mean_precipitation', 'imperviousness', 'min_slope', 'max_slope']
```

Figure 13: Results from Recursive Feature Elimination.

Table 4: Significance of Predictors in the Flood Model.

Predictor	**Significance**	**Rank**
Mean Elevation	Not Significant	5
Mean Slope	Significant	1
Mean Precipitation	Significant	1
Imperviousness	Significant	1
Distance to Drainage	Not Significant	7
Min Elevation	Not Significant	6
Max Elevation	Not Significant	4
Min Slope	Significant	1
Max Slope	Significant	1
Min Precipitation	Not Significant	2
Max Precipitation	Not Significant	3

The table indicates the significance and rank of each predictor in determining flood likelihood. Predictors marked as "Significant" were found to be vital for the model, whereas those labeled as "Not Significant" did not significantly influence the outcome based on recursive feature extraction.

Among the predictors, 'Mean Slope', 'Mean Precipitation', 'Imperviousness', 'Min Slope', and 'Max Slope' were identified as the most influential factors in the prediction model.

4.4 Selected Features

Based on the results from the logistic regression statistics and recursive feature elimination, the features mean slope, mean precipitation, imperviousness, minimum slope, maximum slope, minimum elevation and distance to drainage were chosen as the features that were used to train the model. Histograms showing the distribution of these characteristics is shown in Figures 14 and 15.

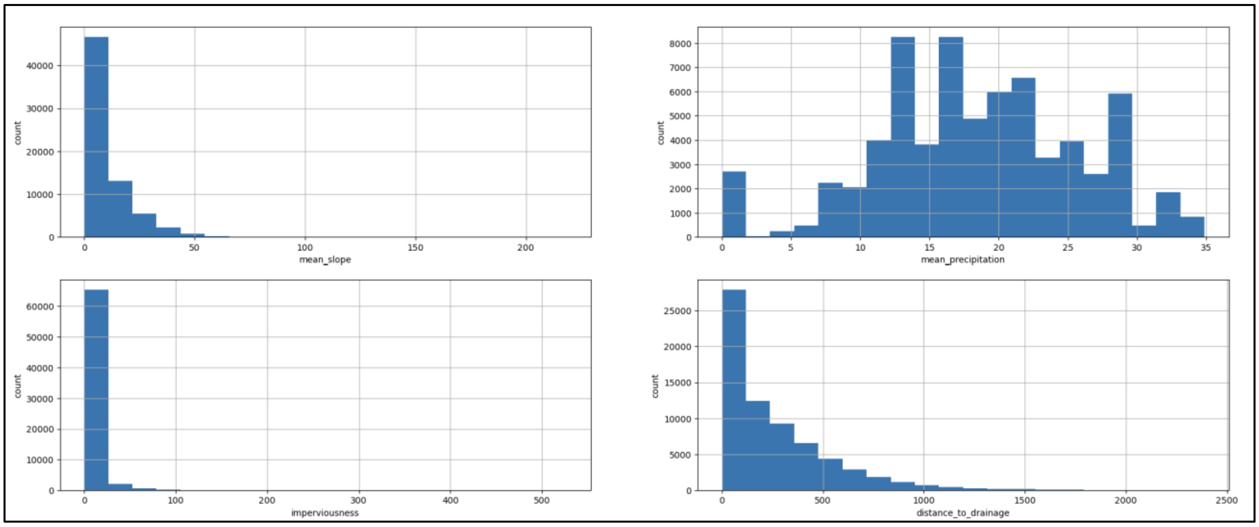

Figure 14: Histograms showing distribution of mean_slope, mean_precipitation, imperviousness and distance_to_drainage.

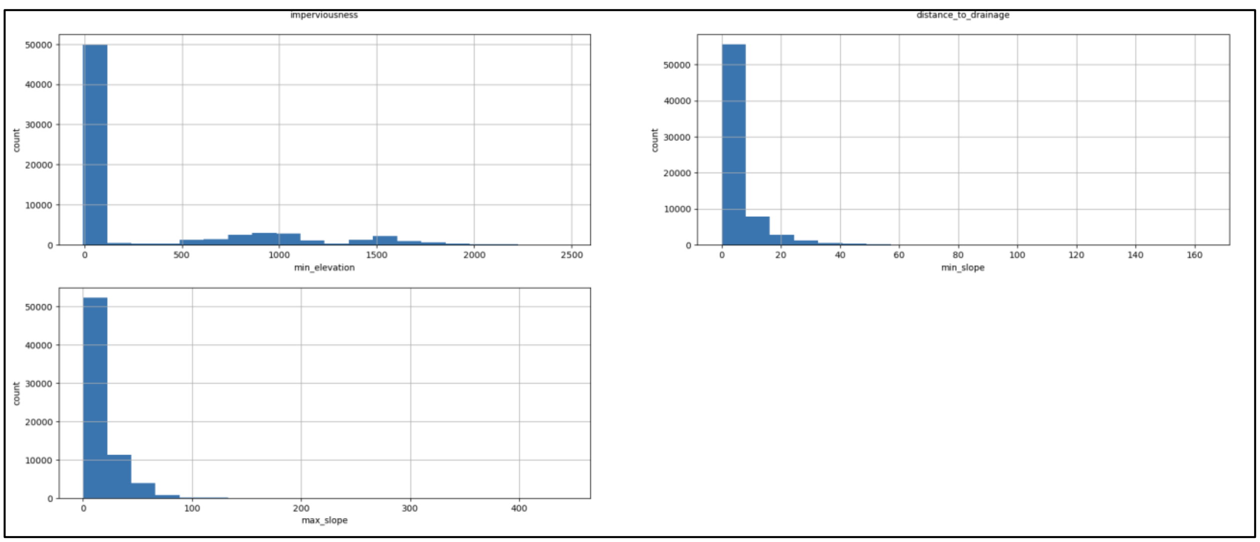

Figure 15: Histograms showing distribution of min_elevation, min_slope and max_slope.

4.5 Model Performance

The dataset was then used to train the model based on Logistic Regression. The data contained 68,404 rows with 8 columns, one of which stored binary values 1 or 0 indicating if the road was flooded or not. But there was a large imbalance in the data with 98.88

percentage of the data being that of roads that were not flooded and with only 1.12 percentage being flooded roads. A chart showing this distribution is shown in Figure 16.

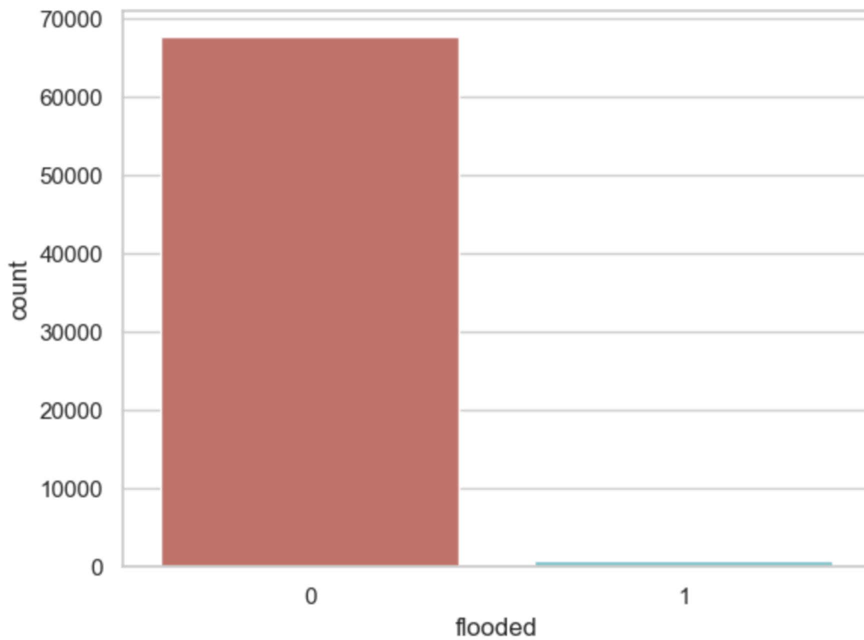

Figure 16: Chart showing the distribution of flooded (1) and non-flooded (0) roads in the dataset.

The mean value of each of the features for flooded and non-flooded roads respectively are shown in Table 5.

Table 5: The mean values of all the characteristics used for training the model.

flooded	mean_slope	mean_precipitation	imperviousness	distance_to_drainage	min_elevation	min_slope	max_slope
0	10.55	18.46	4.95	259.47	299.46	5.42	16.41
1	19.33	16.03	2.05	136.82	511.35	6.09	36.77

To get rid of the data imbalance, the SMOTE algorithm was used. The distribution of the data after this is exactly half of the database having roads that are flooded and the rest non flooded. This is achieved by creating synthetic samples of the minority class which in this case is roads that are flooded. After this, logistic regression model was trained by taking 70 percent of this data. When the model was tested on the rest 30 percent, it gave an accuracy of 0.74. The confusion matrix of the model's performance is shown in Figure 17.

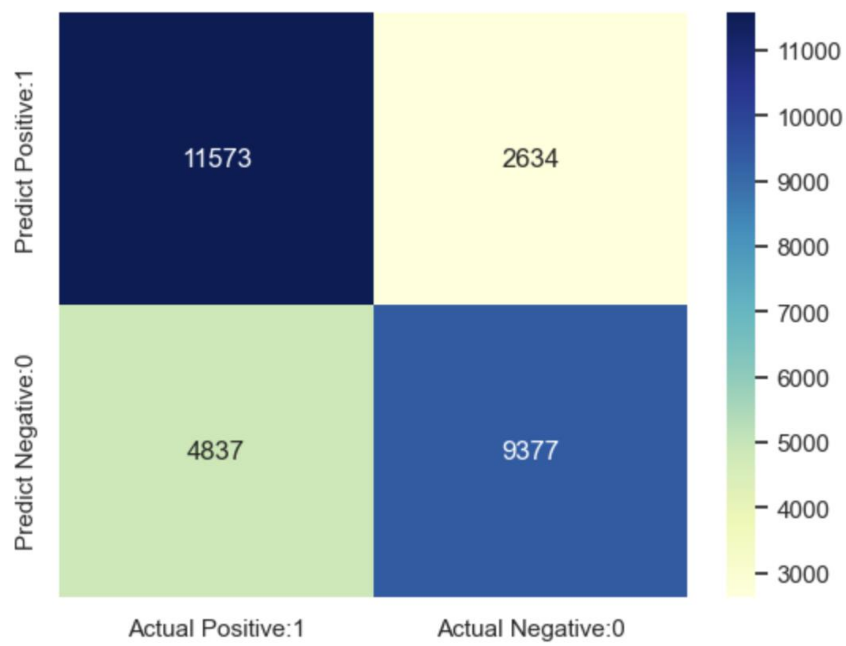

Figure 17: Confusion Matrix of the model's performance.

From Table 6 which are metrics derived from the confusion matrix, it can be seen that for Class 0, the precision is 0.71, recall is 0.81, and the F1-score stands at 0.76. In contrast, for Class 1, precision is reported at 0.78, recall at 0.66, and the F1-score is 0.72. The overall accuracy of the model across both classes is 74%. Both the macro and weighted averages for precision, recall, and F1-score consistently measure at approximately 0.74.

Table 6: Metrics derived from confusion matrix.

	precision	recall	f1-score	support
0	0.71	0.81	0.76	14207
1	0.78	0.66	0.72	14214
accuracy			0.74	28421
macro avg	0.74	0.74	0.74	28421
weighted avg	0.74	0.74	0.74	28421

The Receiver Operating Characteristic (ROC) curve of the model's performance is given in Figure 18.

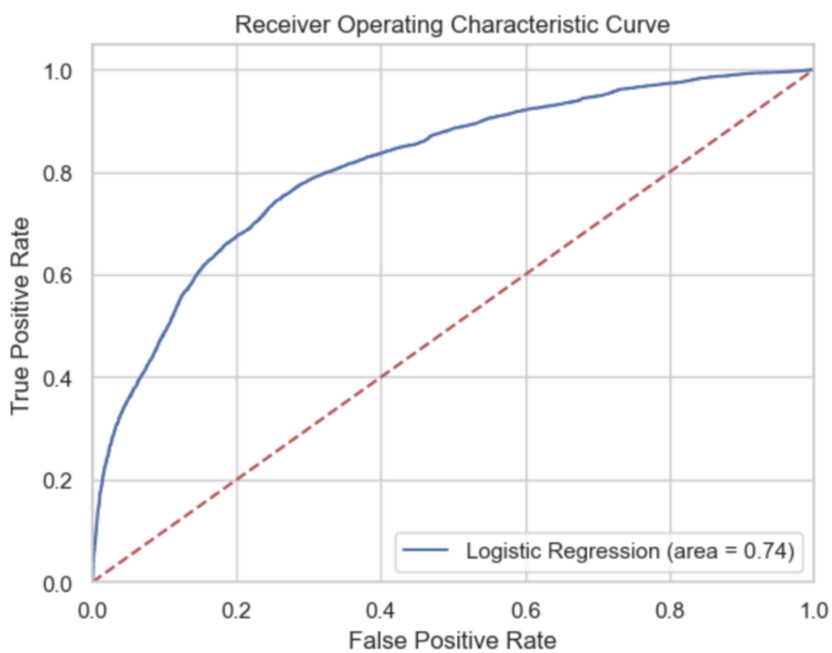

Figure 18: ROC curve of the model's performance.

\

The ROC curve illustrates the performance of a logistic regression model for binary classification. Positioned above the diagonal line, the model demonstrates better predictive power than random guessing, with an Area Under the Curve (AUC) of 0.74, suggesting good but not perfect discrimination between classes. This curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) for various classification thresholds, with the trajectory indicating the model's trade-offs between TPR and FPR. While the model's performance is reasonably good, its adequacy depends on the specific application and context.

Finally, the model was validated using the k-fold cross validation which gave an accuracy of 0.7367 (+/- 0.0046).

5. Discussions

5.1 Vulnerability Assessment of Road Network

Elevation's role in flood vulnerability is pivotal. Typically, lower elevations seem more at risk. In this study, min_elevation and mean_slope have emerged as noteworthy predictors. Specifically, an increase in min_elevation suggests roads at higher elevations are generally less prone to flooding due to the natural flow of water towards lower areas. However, in terrains where higher elevations channel water, rapid runoff can lead to flash floods.

The slope, both min_slope and max_slope, significantly influences flooding dynamics. Roads with a defined slope could encourage faster water runoff, minimizing water accumulation. However, rapid runoff can result in downstream areas flooding, especially if unprepared for such water influxes.

Imperviousness highlights the anthropogenic factor influencing flooding. Urbanization often increases surfaces that prevent water absorption. With rising imperviousness, rainwater struggles to find outlets, boosting the likelihood of flooding. This underscores the importance of permeable materials and efficient drainage in urban planning.

Distance_to_drainage reflects the inherent risks associated with proximity to water bodies or drainage systems. Roads closer to these systems might be at increased risk, especially if the drainage systems are not adequately maintained or are overwhelmed during intense rainfall.

5.2 Creating an Algorithm to Predict Flooding and Feature Importance

The endeavour to harness the power of machine learning in flood prediction reflects a shift from traditional modelling approaches, driven by the need for greater accuracy and adaptability. The logistic regression model, rooted in machine learning, offers a promising alternative by capturing the intricate interplays between various factors.

The initial data imbalance, where a majority of instances represented non-flooded roads, was a significant challenge. In machine learning, a skewed dataset can result in models that are biased towards the majority class. Such models might achieve high accuracy by merely predicting the majority class but fail to capture the nuances of the minority class, which in this case is critical.

Incorporating the SMOTE algorithm was a strategic decision. By generating synthetic instances of the minority class, the algorithm enriched the dataset, allowing for a more balanced training process. This ensures the model is not just statistically sound but also practically relevant.

Furthermore, the application of Recursive Feature Extraction (RFE) provided illuminating insights into feature importance. The results from RFE showcased five significant features: mean_slope, mean_precipitation, imperviousness, min_slope, and max_slope. It's worth noting that while other variables might have intrinsic importance, these five stood out in their predictive capability. This suggests that the slopes (both mean and minimum) and imperviousness play central roles in determining flood vulnerability, reinforcing the earlier discussions. Meanwhile, mean_precipitation emphasizes the direct role of rainfall intensity in flooding events.

5.3 Validating the Algorithm with Data from the 2018 Floods

The catastrophic 2018 Kerala floods offered a real-world context to test the model's robustness. Given the widespread devastation, particularly to the road networks, any model's validation against this backdrop would be stringent.

The model's performance metrics, derived from the confusion matrix, indicated an overall accuracy of 0.74 post-SMOTE. While this signifies the model's robust predictive capabilities, focusing on precision for both Class 0 and Class 1 (0.71 and 0.78 respectively) demonstrates the model's ability to correctly identify the true positive rate, while recall for these classes (0.81 and 0.66 respectively) showcases the model's ability to capture actual positive instances. The F1-score, a harmonic mean of precision and recall, further affirms the model's balanced accuracy for both classes.

Further validation through the k-fold cross-validation yielded a consistent accuracy of 0.7367 with a minor variability of +/- 0.0046, adding another layer of confidence to the model's reliability and stability.

The ROC curve's analysis further complements these insights, showcasing the trade-off between sensitivity (true positive rate) and specificity (true negative rate). The closer the curve is to the top-left corner, the better the model's performance.

While the algorithm has showcased commendable predictive prowess, it's imperative to remember that models, however advanced, have limitations. External factors, data anomalies, or unprecedented climate events can always introduce uncertainties.

6. Conclusions

The pressing global concern of increased flooding events, driven by climate change and human-induced changes to landscapes, underscores the imperative need for effective, predictive measures, especially where urban infrastructure, such as roads, is concerned. The significant aftermath of the 2018 Kerala floods, especially in the Ernakulam and Idukki districts, accentuated this urgency, bringing to light the crippling effects of such catastrophic events on transportation networks and, by extension, on communities and economies.

This study aimed to address the prevalent issue of road network disruption during heavy precipitation events. It ambitiously sought to harness machine learning techniques, notably logistic regression, to predict road disruptions, grounding its research in the severe 2018 floods in Kerala. As delineated in the methodology, the approach was both systematic and holistic, spanning a gamut of steps from comprehensive data acquisition to intricate feature extraction and rectification of data imbalance, before culminating in model training and validation.

The raw data, predominantly sourced from remote sensing products due to scarcity in available data post the 2018 floods, provided a foundation for extracting features pertinent to road networks, such as elevation, slope, permeability, and distance to drainage, among others. Importantly, the data imbalance—a prevalent issue in many natural disaster datasets—was meticulously addressed using the SMOTE method, ensuring that the predictive model would not be overly skewed towards the majority class and would provide a more holistic picture.

The logistic regression model's choice, while rooted in its computational efficiency and capability to provide insights into the relative importance of input variables, also underscored a significant point—the potential of traditional statistical techniques when applied thoughtfully to modern-day challenges. Although the model's pseudo R-squared value and other statistics suggested potential areas of improvement, its insights into significant features were invaluable.

For instance, the importance of features like mean_slope, imperviousness, and distance_to_drainage reinforces the necessity of considering geographical and topographical factors in urban planning and flood mitigation. These features not only affect immediate flood responses but also have long-term impacts on a region's resilience to such events.

Furthermore, the recursive feature extraction provided a refined perspective on the significant predictors, validating some of our initial assumptions while offering newer insights. This iterative process of feature selection underscored the importance of leveraging both statistical and machine learning techniques to refine our predictive capabilities.

However, the broader conclusion from this research transcends its technical achievements. At its core, this study reinforces the importance of proactive, data-driven decision-making in urban planning and disaster response. The ability to predict, with reasonable accuracy, which roads might be disrupted during heavy rainfall can empower local authorities to act in advance, be it in rerouting traffic, reinforcing vulnerable infrastructure, or deploying emergency response teams more effectively.

Furthermore, this study underscores the importance of interdisciplinary collaboration—bridging the gap between traditional hydrological insights, geospatial analysis, and modern data science techniques. In a rapidly changing world, with the spectre of climate change making such events more commonplace, such collaborations are not just beneficial but essential.

Reflecting on the personal motivation behind this study, which emerges from witnessing the devastation of the 2018 Kerala floods firsthand, it's evident that the stakes are high. Ensuring their resilience ensures that a community can withstand, recover, and thrive post-disasters.

While this study provides a blueprint for predicting road disruptions during heavy rainfall in the specific context of Kerala, its methodology and insights can be adapted and applied to other regions, making it a valuable contribution to the broader discourse on urban resilience and flood preparedness. The hope is that the findings of this research will serve as a catalyst for more such predictive initiatives, underpinning global efforts to fortify urban landscapes against the unpredictable wrath of nature.

7. References

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